Production Engineering / Manufacturing Technology- 27K002

PRANAV GURABASAPPA DHODAMANI

Course : **B.Tech**, Manufacturing Science and Engineering, 2026

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CGPA : 7.33

ACADEMIC DETAILS

COURSE	INSTITUTE/COLLEGE	BOARD/UNIVERSITY	SCORE	YEAR
CLASS XII	Dattakala Internation School And Jr College	Maharashtra State Board of Secondary and Higher Secondary Education (MSBSHSE)	87.67 %	2022
CLASS X	VIDYA PRATISHTHAN'S ENGLISH MEDIUM SCHOOL	CBSE	92.2 %	2020

Subjects / Electives	Advanced Joining Technology, Japanese Language, Introduction to AI, Retail
Technical Proficiency	CAD Modelling (SolidWorks/Fusion/Blender), ANSYS Workbench (Structural Analysis), Manufacturing & Fabrication Exposure, Rapid Prototyping (3D Printing), Basic Embedded Systems Integration, Docker, Server Setup, Premiere Pro

PROJECTS

Vectra 1.0 – Custom Ergonomic 3D Navigation Device - Product Development **Jan 2025 - May 2025**

- Led end-to-end product development including design, prototyping, testing, and iterative refinement of an ergonomic 3D navigation device.
- Designed and optimized the mechanical enclosure for durability, ergonomics, and manufacturability using CAD and rapid prototyping
- Integrated embedded hardware and developed firmware for real-time motion sensing and control (<10 ms latency)
- Conducted structured user testing and design iterations based on performance and usability feedback.

Industry CAE-Based Design Optimization Project **Nov 2025 - Dec 2025**

- Performed structural and performance evaluation of an industrial component under real operating boundary conditions using CAE tools.
- Assessed whether the existing design was optimal for load, stress, and deformation requirements.
- Proposed geometry and material modifications to improve performance while considering manufacturing feasibility.
- Bridged design validation and practical engineering constraints, supporting data-driven design improvements.

Tata Elxsi Teliport Season 2 – Development of an automated Inspection system for Railway Bogies - Mechanical & Electronics

Nov 2024 - Jan 2025

- Contributing to the design and development of an automated inspection system for railway bogies
- Working on mechanical integration of sensing systems (depth cameras, proximity sensors) for defect detection
- Supporting development of digital twin-based monitoring to correlate physical inspection with virtual models
- Focused on structural condition monitoring and defect identification in large mechanical assemblies

Home Lab & Self-Hosted Systems Engineering **May 2025 - Jul 2025**

- Designed and maintained a self-hosted server infrastructure, replacing multiple cloud services with containerized open-source solutions
- Gained hands-on experience in Docker-based deployments, networking, storage volumes, and service reliability
- Built automated system workflows using n8n and experimented with locally hosted AI models for edge-style processing
- Developed practical skills in Linux system administration, networking fundamentals, and troubleshooting through live environment management

CERTIFICATIONS

CERTIFICATION	CERTIFYING AUTHORITY
Cyber Security in Manufacturing	University at Buffalo by Coursera
150-hour Robotics and Automation Training Program	SimuSoft Technologies
Intensive Certificate Course in Japanese	SPPU
JLPT N5	Japan Foundation

POSITION OF RESPONSIBILITY

Vice-Captain – Team Velociracers (SAE REEV Series Hybrid vehicle) **Nov 2024 - Present**

- Led manufacturing planning and on-ground implementation of critical vehicle subsystems including suspension mounts, steering assemblies, and structural components
- Responsible for process selection, fabrication sequencing, and tolerance management during chassis and subsystem assembly
- Coordinated with vendors and workshop teams for custom part manufacturing, ensuring parts met functional and fitment requirements
- Resolved real-time assembly and alignment issues, reducing rework and improving build efficiency
- Contributed to system-level integration including regenerative braking implementation
- Took on an initiative of connecting club to now-growing social platform, mainly YouTube, to share, showcase and document the different processes and development stages involved in a student-run car club in the form of weekly vlogs.

COMPETITIONS

Tata Elxsi Teliport Season 2 – Automated Inspection System **Dec, 2024**

- Worked on concepts for automated inspection of railway bogies, focusing on mechanical defect detection and structural condition monitoring
- Gained exposure to real-world maintenance and inspection practices in railway rolling stock systems

Mindspark'24 Ideathon **Oct, 2024**

- Designed a dual-stage actuation piston valve for specified pressure, temperature, and dimensional constraints
- Focused on functional reliability, flow control, and mechanical feasibility

EXTRA CURRICULAR ACTIVITIES

Event

Participated in 95th Punt Formation and joined in as Music Coordinator for the 96th editon.

Accounts Head

Managed club finances, including budgeting, tracking expenses, and preparing financial reports, and also carried out different events on day jobs for 2 of the clubs- BIS Club & Student Welfare Forum

LANGUAGES KNOWN

Japanese, Hindi, Marathi, English